

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

WEB SERVICE TECHNOLOGIES

CASE STUDY 01

RailCo's accounting solution exists as a two-tier client-server application, where the bulk of application logic resides within an executable deployed on client workstations.

Two of the primary tasks this application can perform:

- Enter Customer Purchase Order
- Create Customer Order

The completion of each task involves a series of steps that constitute a business process. This process was originally modeled using standard workflow logic and then implemented as part of a packaged solution. Within the application, the process may or may not be separately represented by individual sets of programming routines. Regardless, it is compiled into a single executable that provides a fixed manner in which the process is automated.

Within a service-oriented business model, the logic behind each process would be partitioned into one or more services. If multiple services are used, the execution of the entire process would involve the composition of these services. In this case, each service may represent a subprocess or even a single step within the process that can be executed independently. For example, the Create Customer Order Process may consist of the following sub-processes:

- Retrieve Purchase Order Data
- Check Inventory Availability
- Generate Back Order
- Publish Customer Order

As we know, a process in its entirety can be viewed and modeled as a service. Additionally, one or more processes can be combined to represent an even larger service. For example, the Create Customer Order and Generate Customer Invoice Processes may be combined to form a single Order Processing Process.

Finally, we would also expect these processes to be flexible so that they can incorporate processes or resources that exist elsewhere in the enterprise. For instance, we may decide to extend the Order Processing Process to include a sub-process that automatically retrieves the customer's current accounts payable mailing address. This sub-process may already exist as part of a separate Customer Contact Reporting Process.

To implement such a model, we need a technical architecture capable of providing the following:

- The ability for business automation logic to be partitioned into units so as to
- properly represent services.
- The ability for these units of logic to be relatively independent of each other so as to support the requirement for them to participate in different compositions.
- The ability for these units of logic to communicate with each other in such a

- manner that their respective independence is preserved.

The fundamental characteristics of service encapsulation, loose-coupling, and messaging, as realized through service-orientation principles and the Web services technology set, collectively fulfill these requirements through the implementation of a primitive SOA.